Does Female Ownership Relate to Productivity–Wage Gaps in Colombia?

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Abstract

In labor markets with frictions such as employer concentration, search costs, and limited worker mobility, firms can exercise monopsony power by setting wages below workers' marginal products, thereby generating productivity-wage gaps. The extent of these gaps depends not only on market structure but also on firm ownership and managerial structure, which shapes wage-setting and employment policies. A growing literature suggests that female leaders place greater weight on worker welfare and are less likely to engage in anticompetitive behavior. This paper examines whether these tendencies translate into reduced monopsony exploitation, specifically whether female ownership attenuates the productivity-wage gap. Using firm-level panel data from the Colombian Annual Manufacturing Survey 2000-2019, I estimate firm-specific productivity-wage gaps following **yeh˙monopsony˙2022<empty citation>** and relate them to the female share of firm ownership in a linear fixed-effects framework controlling for time-varying firm characteristics and industry-level factors. I find that increases in female ownership share are associated with reductions in the productivity-wage gap. These findings carry implications for both equity and efficiency: since wages below marginal product depress employment and distort resource allocation, expanding female ownership in emerging markets may therefore yield meaningful labor market gains.

JEL Classification: D43, D63, J16,J42,J7, M12, M52

Keywords: female ownership, labor market power, productivity-wage gap, labor monopsony, Emerging markets, Colombia

1 Introduction

Across the world, workers are paid substantially less than the value they produce. In Colombia, **amodio˙measuring˙2024<empty citation>** estimates that workers produce approximately 40 percent more than they earn, a gap that represents not only foregone income for millions of workers, but also a significant source of allocative inefficiency for the broader economy. Understanding why this gap exists, and what can be done to reduce it, is a central question in labor economics.

Standard microeconomic theory predicts that in competitive labor markets, wages equal workers' marginal revenue product. Yet a large and growing body of evidence shows that firms are not wage takers: labor markets are characterized by frictions — including search costs, limited job mobility, and employer concentration — that grant firms wage-setting power.[1] When firms exploit this power, they set wages below workers' marginal productivity, generating a productivity-wage gap commonly referred to as a labor markdown. The consequences extend beyond wages: firms with monopsony power also hire fewer workers than is socially optimal, depressing employment and distorting resource allocation across the economy.

Although monopsony power arises from structural features of labor markets, firms differ substantially in how aggressively they exploit it. This variation suggests that firm-level characteristics beyond market structure shape the productivity-wage gap. One such characteristic is firm ownership. A growing literature documents that female leaders tend to place greater weight on worker welfare and employment stability (**matsa˙female˙2013**; **tunyi˙doing˙2023**), interpret productivity signals differently (**flabbi˙female˙2017**; **chen˙why˙2019**), and influence corporate decision-making in ways that may reduce productivity-wage distortions (**levi˙director˙2014**; **flabbi˙female˙2019**). At the same time, labor market power has been shown to affect women more strongly than men (**sharma˙monopsony˙2023**), raising the possibility that female ownership may also attenuate gender-specific productivity-wage distortions within firms. Together, these strands of evidence motivate the central question of this paper: does greater female ownership reduce the extent to which firms exploit monopsony power?

To examine this question, I use firm-level panel data from the Colombian Annual Manufacturing Survey (EAM, from its Spanish initials), collected by Colombia's National Depart-

[1] A growing literature shows that labor markets are often characterized by frictions. See for instance on search costs (**card˙who˙2022**; **azar˙minimum˙2019**; **lentz˙labor˙2010**; **moscarini˙unemployment˙2010**; **manning˙elusive˙2021**), on limited job mobility (**caldwell˙outside˙2024**; **schubert˙employer˙2025**; **starr˙behavioral˙2021**; **mcadams˙non-compete˙2019**), and on employer concentration (**azar˙labor˙2017**; **benmelech˙strong˙2018**; **rinz˙labor˙2022**).

ment of Statistics (**departamento˙administrativo˙nacional˙de˙estadistica˙dane˙encuesta˙nodate**) and covering the period 2000–2019. Colombia provides a particularly informative setting for this analysis. Its manufacturing sector exhibits substantial variation in both female ownership and labor market concentration, and labor market frictions are more pronounced than in high-income economies, making deviations from competitive wage-setting both more likely and more consequential. I estimate firm-specific productivity-wage gaps following the production-function methodology of **yeh˙monopsony˙2022<empty citation>**, which recovers firm-level labor markdowns from output and input data without requiring direct information on market structure. I then relate these estimates to the female share of firm ownership in a linear fixed-effects framework that controls for time-varying firm characteristics and industry-level factors, isolating the within-firm relationship between ownership composition and wage-setting behavior.

The results reveal a robust negative association between female ownership share and the productivity-wage gap: firms with higher female ownership set wages closer to workers' marginal product, consistent with less aggressive exploitation of monopsony power. This finding holds across a range of specifications and is not driven by differences in firm size, productivity, or industry composition. These results contribute to two distinct literatures. First, they add to the growing body of work on the determinants of labor market power by showing that ownership structure shapes the productivity-wage gap. Second, they extend the literature on female leadership and firm behavior by connecting gender composition in ownership to a concrete labor market outcome with direct welfare implications.

These findings carry important policy implications, particularly for emerging markets where labor market frictions are often more pronounced and monopsony rents are larger. Policies that expand female ownership through access to finance, legal protections, or targeted investment incentives may generate returns that extend beyond the firms themselves, improving wages and employment for workers more broadly. More generally, the results suggest that the distributional consequences of labor market power are not determined by market structure alone, and that who owns firms matters for how those firms treat their workers.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature and estimations on the productivity-wage gap. Section 3 presents a theoretical framework that illustrates how monopsony power generates a productivity-wage gap and how this gap can be estimated using firm-level data. Section 4 describes the data and presents summary statistics. Section 5 outlines the empirical strategy. Section 6 presents the main results and robustness checks. Finally, section 7 provides the conclusion, policy implications, and directions for future research.

2 Literature Review

Table 1: Summary of Related Literature

Paper	Topic	Data	Main Finding
De Loecker, Eeckhout, & Unger (2020)	Labor market concentration	U.S. manufacturing	Production approach for a robust estimation of elasticities. Findings suggest markups rise not just to cover higher fixed costs, but also due to firms' market power.
Yeh, Macaluso & Hershbein (2022)	Labor markdown estimation	U.S. manufacturing	Most manufacturing plants operate in a monopsonistic environment. Average markdown 1.53 workers earning only 65 cents on the marginal dollar generated.
Amodio & de Roux (2024)	Colombia Labor market power	Colombian manufacturing plants (EAM)	Estimate a firm-level labor supply elasticity of around 2.5, implying that workers produce about 40% more than their wage level
Inter-American Development Bank (2025)	Latin-America and the Caribbean Market Power	Firm-Data LAC (For Colombia EAM)	On average, workers in Latin America and the Caribbean take home in the form of wages 50 percent of what they contribute to their employers.
Matsa & Miller (2013)	Female leadership	Nordic companies (Norway's 2006 quota)	Female leaders tend to emphasize employment stability and worker retention.
Flabbi et al. (2019)	Female executives	Italy employer-employee panel data	Female leadership influences wage-setting and reduce gender gaps within firms.
Levi, Li & Zhang (2014)	Female executives	Aquisition bids by S&P 1500 companies	Female directors less overestimate merger gains and help create shareholder value

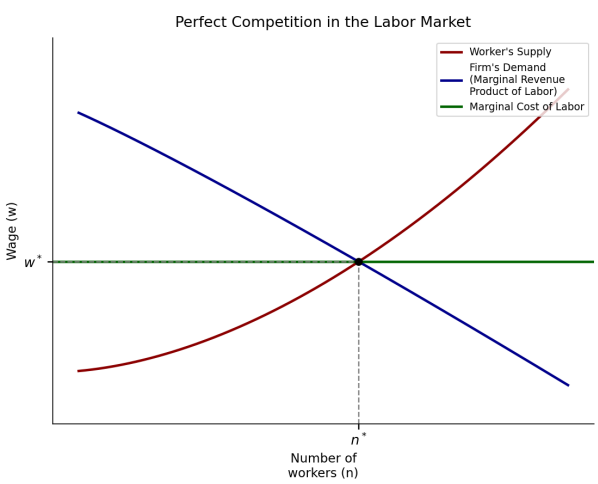
3 Theoretical Framework

To understand the gap between wages and workers' marginal productivity, a useful starting point is the benchmark case of a competitive labor market. As shown in Panel (a) of Figure 1, under perfect competition the marginal cost of labor is constant and equal to the market wage, which each firm takes as given. Firms therefore hire labor up to the point where the marginal revenue product of labor (MRPL) equals the wage.

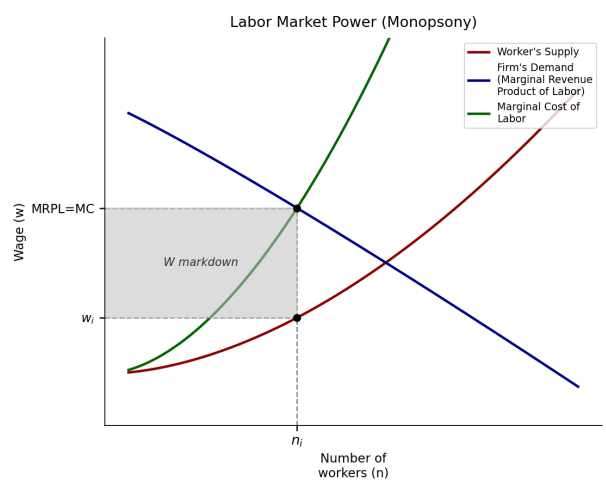
Monopsony power can be exercised by any employer who faces an upward-sloping labor supply curve, as shown in Panel (b) of Figure 1. Hiring an additional worker requires raising the wage not only for the marginal worker but, when wages are set uniformly, for all inframarginal workers as well. As a result, the marginal cost of labor lies strictly above the labor supply curve. The profit-maximizing firm sets employment where MRPL = MCL, yielding both lower employment and a lower wage relative to the competitive benchmark. Crucially, at the optimum, workers are paid a wage w_i determined by the labor supply curve, yet their marginal contribution to firm revenue satisfies MRPL $> w_i$. The wedge between the marginal revenue product of labor and the wage is known as the *markdown*, illustrated by the shaded region in Panel (b), and it is the central object of interest in this paper.

Figure 1: Labor Markets

(a) Perfect Competition



(b) Monopsony



To see how this markdown arises formally, following **ashenfelter˙labor˙2010<empty citation>** consider a firm's profit maximization problem with labor as the only input. Under perfect competition, the firm takes the market wage $\bar{w}$ as given and chooses labor input L to maximize profits:

$$\max_L \Pi = R(L) - \bar{w}L, \tag{1}$$

Where $R(L) \equiv P * Q(L)$ is the firm's revenue as a function of labor input. The first-order condition for profit maximization equates the marginal revenue with the marginal cost such that:

$$R'(L) = \bar{w} \tag{2}$$

Under monopsony, the firm faces an upward-sloping labor supply curve, so the wage is an increasing function of employment $w(L)$, as shown in figure 1. The profit maximization problem becomes:

$$\max_L \Pi = R(L) - w(L)L, \tag{3}$$

with first-order condition:

$$R'(L) = w(L) + w'(L)L \tag{4}$$

Note that the monopolistic firms also maximizes profits by equating marginal revenue and marginal cost, but the marginal cost of labor is greater than the wage in this case. The right-hand side of the equation exceeds the wage $w(L)$ by the term $w'(L)L$ which captures the additional cost of raising wages for all inframarginal workers when hiring one more.

Rearranging yields to the Lerner index formula for the labor market:

$$\frac{R'(L) - w(L)}{w(L)} = \frac{w'(L)L}{w(L)} \equiv \varepsilon_L^{-1} \tag{5}$$

Where $\varepsilon_L^{-1} \equiv \frac{w'(L)L}{w(L)}$ is the firm's perceived inverse elasticity of labor supply, capturing how much the wage must increase to attract an additional worker. This expression shows that the percentage gap between the marginal revenue product of labor and the wage equals the inverse of the firm's perceived labor supply elasticity: the more inelastic the labor supply (lower ε_L), the larger the gap between the marginal revenue product of labor and the wage.

Following **yeh˙monopsony˙2022<empty citation>**, I define the labor markdown ν as the ratio of the marginal revenue product of labor to the wage:

$$\nu \equiv \frac{R'(L)}{w(L)} = \varepsilon_L^{-1} + 1 \tag{6}$$

Under this definition, $\nu = 1$ under perfect competition (infinite labor supply elasticity)

and $\nu > 1$ whenever labor supply is upward-sloping. It is therefore sufficient to estimate the firm's perceived labor supply elasticity ε_L to recover the markdown ν. However, as highlighted by **yeh˙monopsony˙2022<empty citation>**, estimating ε_L in a general setting is challenging, as it requires disentangling the effects of labor market power from those of output market power (monopoly) and from other confounding factors. To address this challenge, I adopt the production approach of **de˙loecker˙rise˙2020<empty citation>**, which allows for the estimation of firm-level markdowns using data on output and input costs without requiring direct information on market structure.

3.1 Markdown calculation - Production Approach

To understand the production approach from **de˙loecker˙rise˙2020<empty citation>** let's consider a simple Cobb-Douglas production function with three inputs: labor L, capital K, and materials M, where L and M adjust frictionlessly within a period, while capital is subject to adjustment frictions. I removed the subscript it for simplicity, but all variables are indexed by time and firm.

The approach relies on the assumption that materials (M) is a fully flexible input satisfying three conditions: it incurs no adjustment costs within a period; its price is taken as given by the firm; and it is chosen statically, in the same period in which output is produced.

The firm's cost minimization problem is:

$$\mathcal{L} = rK + w(L)L + p^m M - \lambda \left(P\bar{Q} - AK^{\theta^K} L^{\theta^L} M^{\theta^M} \right) \tag{7}$$

where θ^K, θ^L, θ^M are the output elasticities of capital, labor, and materials, respectively and A denotes firm-specific productivity. The price of capital is r, the wage function is $w(L)$, and the price of materials is p^m. P is the price of output, $\bar{Q}$ is the quantity of output produced, and λ is the Lagrange multiplier associated with the production constraint which captures the marginal cost of production

The first-order conditions for M and L are:

$$\frac{\partial \mathcal{L}}{\partial M} = p^m - \lambda \theta^M A K^{\theta^K} L^{\theta^L} M^{\theta^M - 1} = 0. \tag{8}$$

$$\frac{\partial \mathcal{L}}{\partial L} = w'(L)L + w(L) - \lambda \theta^L A K^{\theta^K} L^{\theta^L - 1} M^{\theta^M} = 0, \tag{9}$$

We can simplify the first-order conditions as:

$$p^m = \lambda \theta^M \frac{Q}{M} \tag{10}$$

$$w'(L)L + w(L) = \lambda\theta^L \frac{Q}{L} \tag{11}$$

The Lagrange multiplier λ captures the marginal cost of production, thus, the firm's markup, defined as the price-to-marginal-cost ratio is therefore:

$$\mu = \frac{P}{\lambda}. \tag{12}$$

Substituting and rearranging in the first-order conditions for materials:

$$p^m = \theta^M \frac{PQ}{M\mu}, \tag{13}$$

We obtain an expression for the markup in terms of observable quantities:

$$\mu = \theta^M \frac{PQ}{p^m M}, \tag{14}$$

For the first order condition for labor, let's first express the left-hand side in terms of the labor supply elasticity $w'(L)L + w(L) \equiv \frac{w'(L)L}{w(L)} + 1 \equiv (\varepsilon_L^{-1} + 1)w(L)$. Rearranging the first-order condition for labor gives:

$$(\varepsilon_L^{-1} + 1)w(L) = \frac{\theta^L}{\mu}\frac{PQ}{L}, \tag{15}$$

Now, we know from the theory section that the labor markdown ν is defined as the ratio of the marginal revenue product of labor to the wage, which can be expressed as:

$$\nu \equiv \frac{\text{MRPL}}{w(L)} = \varepsilon_L^{-1} + 1. \tag{16}$$

Substituting $\varepsilon_L^{-1} + 1 = \nu$ into the first-order condition for labor gives:

$$\nu = \frac{\theta^L}{\mu}\frac{PQ}{Lw(L)}. \tag{17}$$

Substituting the expression for the markup μ derived from the first-order condition for materials into the above equation for markdown ν yields:

$$\nu = \frac{\theta^L}{\theta^M} \cdot \frac{p^m M}{w(L)L}. \tag{18}$$

This expression shows that the markdown (productivity-wage gap) can be estimated using the ratio of the firm's expenditure on materials to its wage bill, scaled by the ratio of output elasticities. The output elasticities θ^L and θ^M are estimated from the production

function, while the input expenditures are computed from observed data on wages, materials, output, and prices. This approach allows for the estimation of firm-level markdowns while controlling for both output and labor market power.

4 Data

This paper uses firm-level data from the Annual Manufacturing Survey (*Encuesta Anual Manufacturera*, EAM) conducted by Colombia's National Administrative Department of Statistics (DANE). The EAM is a census of medium and large manufacturing establishments that provides detailed and standardized information on firms' production, input use, employment composition, and financial accounts. I restrict the analysis to the period 2000–2019 to ensure data consistency and to avoid disruptions associated with the COVID-19 pandemic. Although the data are collected at the establishment level, I aggregate to the firm level by summing across all plants belonging to the same firm in a given year, using the unique firm identifier provided in the survey. This aggregation is done to capture the full scope of each firm's operations and to align with the theoretical framework, which is based on firm-level decision-making.

Administrative firm-level data frequently contain extreme values arising from measurement error, typing mistakes, or genuine but highly influential outliers. Left uncorrected, such values can severely bias the production-function estimates that underpin the labor market power calculations. Following **yeh˙monopsony˙2022**, I apply three filters. First, I winsorize the key variables: output, labor, capital, and materials, at the top and bottom 1 percent of their respective distributions to mitigate the influence of extreme observations. Second, I exclude observations with non-positive values for output or any input, as these are inconsistent with the production framework. Third, to ensure that materials satisfy the flexible-input assumptions required for identification, I compute each firm's materials cost share, defined as materials expenditure as a share of total costs, and exclude observations whose materials cost share falls below one quarter of the sector-year median. Firms with anomalously low materials cost shares are unlikely to satisfy the conditions of no adjustment costs, static input choice, and the absence of input market power, and their inclusion could contaminate the proxy-variable estimator. The final dataset is an unbalanced panel of 6,323 Colombian manufacturing firms observed over 2000–2019, yielding 29,942 firm-year observations spanning 92 industries and 23 geographic regions.

Female ownership. The primary explanatory variable is an indicator equal to 1 if the firm has at least one female owner and 0 otherwise. This variable is constructed from

survey items recording the number of male owners (`c4r4c1t`) and female owners (`c4r4c2t`).[2] Table 2 shows that 53.4 percent of firm-year observations have at least one female owner, at least one female owner is common in Colombian manufacturing, albeit with substantial variation across firms and over time. As a continuous complement, I also construct the female ownership share, defined as the number of female owners divided by the total number of owners. The mean share is 35.3 percent with a standard deviation of 37.6 percent. The wide dispersion and near-zero skew of this distribution motivate the use of the binary indicator as the primary measure, which avoids imposing a linear relationship between the productivity-wage gap and the ownership share.[3].

In terms of the other firm characteristics, firms employ an average of 34 workers, with annual wages per employee averaging $1.83 thousand USD and mean gross output of $1.22 million USD. The average age of firms is 4.6 years; all of these variables exhibit substantial dispersion, reflecting the heterogeneity in firm size and performance that is typical of manufacturing surveys of this type.

Table 2: Descriptive Statistics

Variable	*Mean*	*Min*	*Max*	*Std. Dev.*
Binary 1 if $\geq$ one female owner	0.535	0	1	0.499
Female ownership share (%)	35.319	0	100	37.579
Employees (N)	34	1	4,025	99
Wage per employee ($ thousands USD)	1.827	91	349.180	2.296
Gross output ($ millions USD)	1.219	536.010	756.959	11.356
Firm age (years)	4.628	1	20	3.696

Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019. Note: The wage variable corresponds to the total wage bill of the firm divided by the number of employees, while gross output is proxied by total sales at factory price without direct taxes. To simplify the presentation, all dollar values are calculated using an exchange rate of 1 USD = 3,670 COP.

[2]These variables represent the total number of owners, partners, and unpaid family members, disaggregated by gender and occupational category—professionals; technicians and technologists; production workers (foreign and local); operators and producers; and managers and sales personnel. I use the sum across all occupational categories to construct the total ownership count.

[3]One important limitation of both ownership measures is that total ownership counts are reported for only approximately 30 percent of firm-year observations, which may reflect genuine non-reporting or survey design issues. The analysis is restricted to observations with nonzero ownership counts, which introduces a potential selection concern: firms for which ownership data are observed may differ systematically from those for which they are not. The results should therefore be interpreted with appropriate caution.

4.1 Markdown Estimation

The main outcome variable is the firm-level labor markdown, which captures the gap between a worker's marginal revenue product and their wage. Following **yeh˙monopsony˙2022<empty citation>** the markdown can be expressed in terms of observable quantities as:

$$\nu_{it} = \frac{\theta_{it}^{L}}{\theta_{it}^{M}} \cdot \frac{p_{it}^{m} M_{it}}{w_{it} L_{it}}, \tag{19}$$

where $p_{it}^{m} M_{it}$ is total expenditure on intermediate inputs and $w_{it} L_{it}$ is the total wage bill, both observed directly in the data. To recover firm-year output elasticities θ_{it}^{L} and θ_{it}^{M}, I estimate a translog gross-output production function following **de˙loecker˙rise˙2020<empty citation>**:

The values $p^m M$ and $w(L)L$ can be extracted directly from the dataset, $p^m M$ is the total expenditure on raw materials, and $w(L)L$ is the total wage bill which corresponds to the total remuneration paid to permanent and temporary employees, both directly hired and outsourced. To estimate θ^L and θ^M, I use **de˙loecker˙rise˙2020<empty citation>** code which uses a translog production function with polynomial terms as follows:

$$\begin{aligned} \log(GO)_{it} = \beta_0 + \beta_1 \log(K)_{it} + \beta_2 \log(L)_{it} + \beta_3 \log(M)_{it} \\ + \beta_4 (\log K_{it})^2 + \beta_5 (\log L_{it})^2 + \beta_6 (\log M_{it})^2 + \beta_7 \log(K)_{it} \log(L)_{it} \\ + \beta_8 \log(K)_{it} \log(M)_{it} + \beta_9 \log(L)_{it} \log(M)_{it} + \gamma_t + \varepsilon_{it}. \end{aligned} \tag{20}$$

where GO_{it} is gross output, K_{it} is capital, L_{it} is labor, M_{it} is intermediate inputs, and γ_t are year fixed effects. Labor is measured as the total number of employees. Capital is proxied by the book value of fixed assets, with its rental cost computed via a user-cost formula that incorporates asset-specific depreciation rates from **de˙loecker˙rise˙2020<empty citation>** and year-specific interest rates and inflation from the Central Bank of Colombia (**banco˙de˙la˙republica˙de**
Intermediate inputs are measured as total expenditure on raw materials, intermediate goods, and packaging; consistent with the requirements of the proxy-variable estimator, materials serve as the flexible input in production. Gross output is proxied by firm-level total sales at factory prices, net of direct taxes, as reported in the Colombian Manufacturing Survey (EAM).

The production function is estimated separately for each 4-digit ISIC sector via the two-stage GMM procedure of **de˙loecker˙rise˙2020<empty citation>**, which corrects for the endogeneity of input choices.[4] Given the translog structure, output elasticities are firm-year

[4]Because firms observe their own productivity before choosing inputs, OLS estimates are biased upward. The estimator recovers a productivity proxy $\hat{\phi}_{it}$ from a first-stage OLS regression, then constructs the productivity shock $\omega_{it}(\beta) = \hat{\phi}_{it} - \mathbf{x}_{it}'\beta$ for a candidate parameter vector β. Assuming a Markov process for productivity, its conditional expectation is approximated by a cubic polynomial in ω_{it-1}. The innovation $\xi_{it} = \omega_{it} - E[\omega_{it} \mid \omega_{it-1}]$ is orthogonal to lagged inputs by assumption, providing the GMM moment condi-

specific and recovered as the partial derivatives of output with respect to each input:

$$\theta_{K,it} = \beta_1 + 2\beta_4 \log(K)_{it} + \beta_7 \log(L)_{it} + \beta_8 \log(M)_{it}, \tag{21}$$

$$\begin{aligned} \theta_{L,it} &= \beta_2 + 2\beta_5 \log(L)_{it} + \beta_7 \log(K)_{it} + \beta_9 \log(M)_{it}, \\ \theta_{M,it} &= \beta_3 + 2\beta_6 \log(M)_{it} + \beta_8 \log(K)_{it} + \beta_9 \log(L)_{it}. \end{aligned} \tag{22}$$

For sectors where the translog GMM fails to converge or yields non-positive elasticities, I fall back to a Cobb-Douglas specification estimated via the procedure of **ackerberg˙identification˙2015<e** which imposes sector-constant elasticities but is considerably more robust to limited panel variation.[5]

Table 3 reports the distribution of estimated outputelasticities, markups, and markdowns. The estimated output elasticities are broadly consistent with the existing literature and lend credibility to the markdown estimates. The mean output elasticity of labor is 0.37 and the mean elasticity of materials is 0.62, with capital averaging 0.08 — a pattern consistent with the capital-light, assembly-intensive nature of much Colombian manufacturing and in line with estimates reported by **de˙loecker˙rise˙2020<empty citation>** and **yeh˙monopsony˙2022<empty citation>** for U.S. manufacturing using the same methodology. The dominance of materials in the cost structure further supports their use as the flexible input in the proxy-variable estimator.

The mean product markup is 1.23, implying that the average firm prices output approximately 23 percent above marginal cost. The median of 1.31 falls within the range estimated by **de˙loecker˙rise˙2020<empty citation>** for U.S. manufacturing, and is considerably below the average of 87 percent above marginal cost reported for the broader Latin American and Caribbean region by **inter-american˙development˙bank˙competelac˙2025<empty citation>**.

Turning to the markdown estimates, the median firm-level markdown is 1.93, implying that the median worker generates nearly twice as much revenue at the margin as they take home in wages — equivalently, for every dollar generated at the margin, the worker receives approximately 52 cents. This exceeds the median of 1.53 estimated for U.S. manufacturing by **yeh˙monopsony˙2022<empty citation>**, consistent with stronger monopsony power in the Colombian labor market. The estimate is also broadly aligned with

tions. Capital is treated as predetermined and enters in current values; labor and materials are instrumented with their lagged values and lagged polynomial terms. The GMM criterion is minimized using a Nelder-Mead optimizer, with a gradient-based Rvmmin fallback when convergence is not achieved.

[5]The Cobb-Douglas specification takes the form $\log(GO)_{it} = \beta_0 + \beta_K \log(K)_{it} + \beta_L \log(L)_{it} + \beta_M \log(M)_{it} + \gamma_t + \varepsilon_{it}$, estimated with the same two-stage GMM logic. Because elasticities are sector-constant under Cobb-Douglas, I impute the sector-level estimates to all firms in that sector. The vast majority of firm-year observations are covered by the translog estimator.

amodio˙measuring˙2024<empty citation>, who report a labor supply elasticity of approximately 2.5 for Colombia, implying workers produce around 40 percent more than their wage. The substantial dispersion in markdowns — the interquartile range runs from 1.04 to 3.41 — mirrors the firm-level heterogeneity documented in the literature (**yeh˙monopsony˙2022**) and motivates the regression analysis in Section 5.

Table 3: Production Function Parameters

Variable	*Mean*	*p25*	*Median*	*p75*
$\theta_{L,it}$ Output elasticity of labor	0.368	0.252	0.389	0.433
$\theta_{K,it}$ Output elasticity of capital	0.082	0.035	0.065	0.108
$\theta_{M,it}$ Output elasticity of materials	0.623	0.526	0.641	0.700
μ_{it} Markup	1.231	1.026	1.310	1.736
ν_{it} **Markdown (productivity-wage gap)**	**1.059**	**1.028**	**1.906**	**3.367**

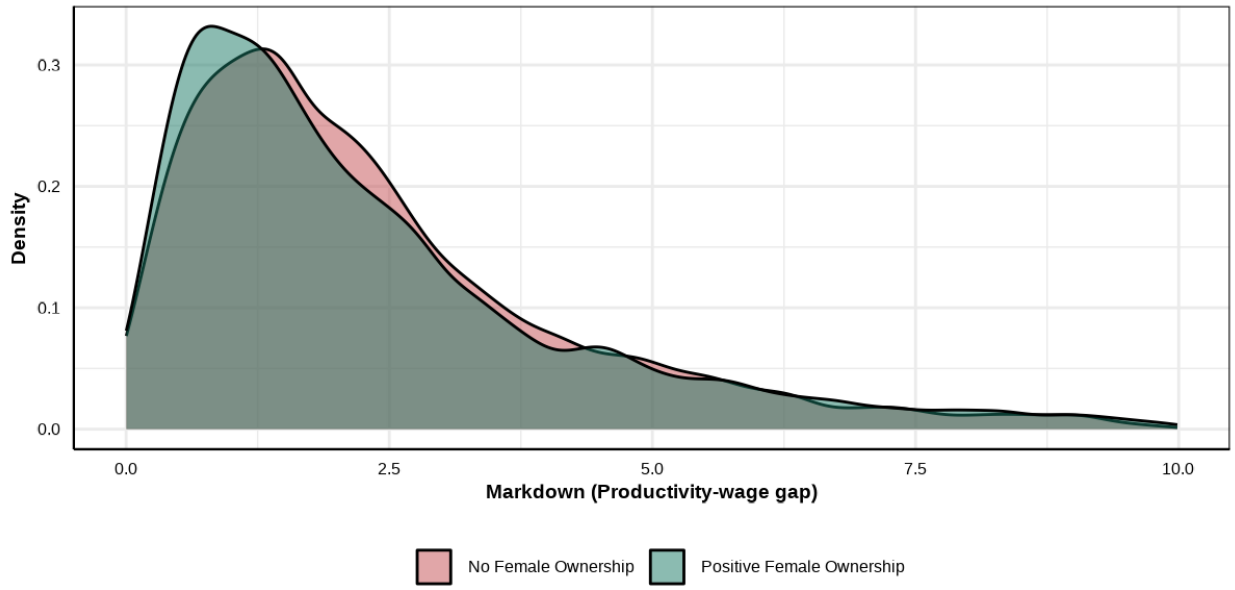
Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019. Note: Output elasticities are estimated via the translog production function approach of **de˙loecker˙rise˙2020**, with a Cobb-Douglas fallback following **ackerberg˙identification˙2015** for sectors where the translog GMM does not converge. Markups and markdowns follow **yeh˙monopsony˙2022**.

4.2 Descriptive Analysis

Figure 2 plots the distribution of the markdown (productivity-wage gap) separately for firms with no female ownership and firms with at least partial female ownership. Two features stand out. First, firms with no female ownership exhibit a distributionshifted toward higher markdown values, indicating that workers' marginal revenue product is further above their wage relative to firms with at least one female owner. Second, the distribution for partially or fully female-owned firms is more concentrated near one — the competitive benchmark — suggesting that female ownership is associated with a smaller wedge between productivity and wages.

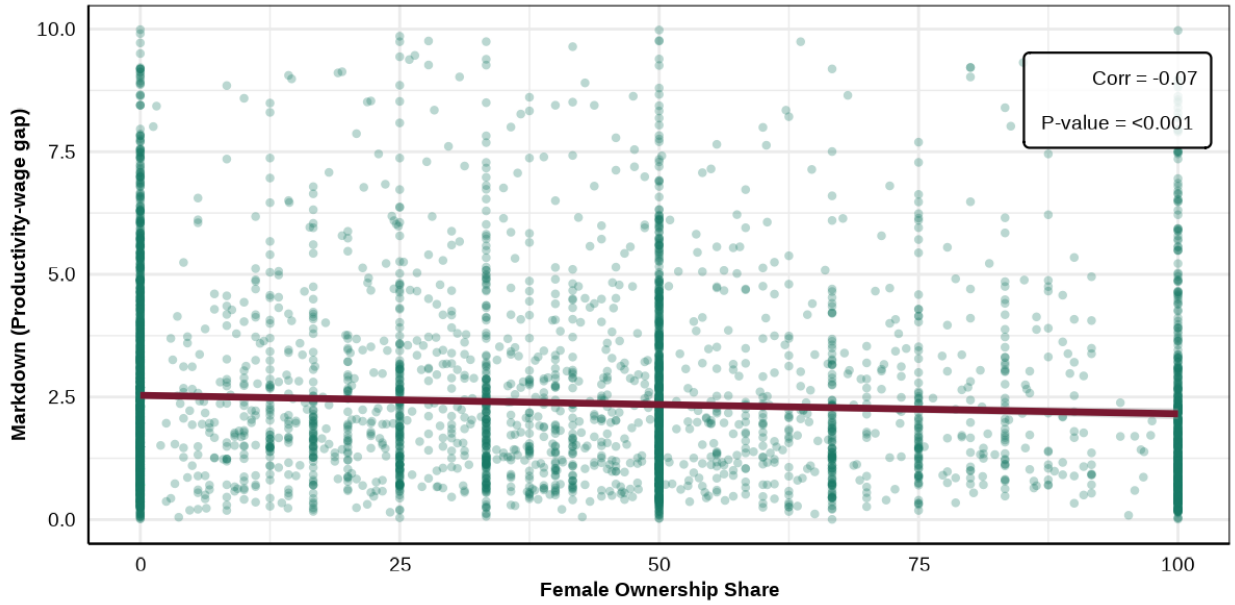
This pattern is corroborated by Figure 3, which plots the markdown against the female share of ownership at the firm level. The correlation is negative ($\rho = -0.07$, $p < 0.001$), providing preliminary evidence that greater female ownership is associated with workers capturing a larger share of their marginal revenue product. While the correlation is modest in magnitude, it is precisely estimated and its direction is stable across the distribution, motivating the more rigorous regression analysis that follows.

Figure 2: Distribution of Markdown by Female Ownership



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019.

Figure 3: Markdown and Female Share of Ownership



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019.

5 Empirical Framework

To examine the relationship between female ownership and the productivity-wage gap, I estimate the following linear regression:

$$\nu_{it} = \alpha + \beta \, \text{Female}_{it} + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_t + \alpha_s + \alpha_r + \varepsilon_{it}, \tag{23}$$

where ν_{it} is the labor markdown for firm i in year t, as estimated in Section **??**. The coefficient of interest is β, which I estimate under two alternative definitions of Female_{it}. In the first, Female_{it} is a binary indicator equal to one if the firm has at least one female owner, so that β captures the average difference in the markdown between partially or fully female-owned firms and firms with no female ownership, conditional on controls and fixed effects. In the second, Female_{it} is the continuous female share of ownership, so that β captures the change in the markdown associated with a one percentage-point increase in female ownership share. A negative estimate of β in either specification would indicate that female ownership is associated with workers capturing a larger share of their marginal revenue product.

The vector $\mathbf{X}_{it}$ includes firm-level controls for age, employment, sales, and total factor productivity recovered from the production function estimation described in Section **??**. Firm age accounts for the possibility that older firms accumulate labor market power through reputation and search frictions (**yeh˙monopsony˙2022**). Employment and sales control for scale effects on both production technology and wage-setting capacity. Productivity enters as a control because more productive firms tend to generate larger markups and markdowns, and because **moreno-gomez˙relationship˙2018<empty citation>** finds that female-owned firms in Colombia are systematically more productive, omitting it would risk confounding the ownership effect with a productivity channel.

The specification includes three sets of fixed effects. Year fixed effects α_t absorb common macroeconomic shocks and aggregate trends in markdowns over time, including business-cycle variation in labor market tightness. Sector fixed effects α_s account for permanent differences across industries in production technology, product market concentration, and the degree of labor market competitiveness. Region fixed effects α_r control for local labor market conditions, including geographic variation in workers' outside options and commuting costs. Standard errors are clustered at the firm level to account for serial correlation in the markdown within firms over time.

6 Results

Table 4: Regression Results

	(1)	(2)	(3)
`own_w_share`	0.066***	0.010*	0.012**
	(0.007)	(0.006)	(0.006)
Fixed Effects	Year	Year-Industry	Year-Industry & Geography

Note: *p<0.10; **p<0.05; ***p<0.01. Standard errors in parentheses.

Table 4 presents the results of the regression analysis. In the baseline specification (Column 1), which includes only year fixed effects, the coefficient on the female ownership share is positive and statistically significant at the 1 percent level, suggesting that a higher share of female ownership is associated with a higher inverse markdown, or equivalently, a lower degree of monopsony power. Specifically, a 10 percentage point increase in the share of female ownership is associated with a 0.66 percentage point increase in the inverse markdown, implying that for every 100 dollars of marginal revenue product, workers receive approximately 0.66 dollars more in wages.

When industry fixed effects are added in Column 2, the coefficient on female ownership share remains positive but its magnitude decreases substantially to 0.010 and is only significant at the 10 percent level. This suggests that part of the association observed in the baseline may be driven by industry composition, as certain industries with higher female ownership may also have more competitive labor markets. Finally, in Column 3, which includes both industry and geography fixed effects, the coefficient on female ownership share remains positive and becomes statistically significant at the 5 percent level, with a magnitude of 0.012. This indicates that even after controlling for both industry and regional factors, there is still a positive association between female ownership and the inverse markdown, although the effect size is smaller than in the baseline specification.

7 Conclusion

to be done